Strip-Plot Designs

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Introduction

Strip-plot designs — also known as split-block designs — apply two hard-to-change factors in perpendicular strips across a block, with the resulting factor combinations realised in the cells of the cross. They arise naturally in agricultural field experiments where one factor (e.g., irrigation level) is applied along rows and another (e.g., fertiliser type) along columns, in industrial manufacturing where two stages of processing each cover an entire pallet, and in food and pharmaceutical settings where two restrictions on randomisation operate in different directions. The defining feature is three independent error strata — one for the row factor, one for the column factor, and one for their interaction — each requiring its own appropriate F-test denominator.

Prerequisites

A working understanding of split-plot designs, the concept of multiple error strata, and mixed-effects model specification with nested random effects.

Theory

The strip-plot model has three error strata corresponding to the three sources of restricted randomisation:

- **Row factor**: applied to entire rows; tested against row-stratum error (variation among rows within blocks).
- **Column factor**: applied to entire columns; tested against column-stratum error.
- **Row $\times$ column interaction**: tested against the cell-level (smallest) error.

Each main effect therefore has lower power than it would in a completely randomised design, because its denominator is built from a coarser stratum, but the design is operationally feasible when full randomisation of the two hard-to-change factors is impractical or prohibitively expensive.

Assumptions

The three error strata are independent, errors are Normal and homoscedastic at each stratum, and the random-effects structure is correctly specified — typically random effects for row-within-block and column-within-block, with the cell-level residual capturing the interaction-level error.

R Implementation

```
library(lme4); library(lmerTest)

set.seed(2026)
n_rep <- 3
rows <- factor(rep(1:3, each = 4, times = n_rep))
cols <- factor(rep(rep(1:4, 3), n_rep))
rep_f <- factor(rep(1:n_rep, each = 12))
df <- data.frame(rep = rep_f, rows, cols)
```

```r
df$row_batch <- interaction(df$rep, df$rows)
df$col_batch <- interaction(df$rep, df$cols)

row_re <- rnorm(nlevels(df$row_batch), 0, 0.5)[df$row_batch]
col_re <- rnorm(nlevels(df$col_batch), 0, 0.4)[df$col_batch]

df$y <- 10 + 0.6 * as.numeric(df$rows) + 0.4 * as.numeric(df$cols) +
        row_re + col_re + rnorm(nrow(df), 0, 0.3)

fit <- lmer(y ~ rows * cols + (1 | row_batch) + (1 | col_batch),
            data = df)
anova(fit)
```

Output & Results

`lmer()` with two crossed random-effect terms (row-within-replicate and column-within-replicate) and the cell-level residual recovers the three error strata. `anova()` returns the appropriate F-tests for the row and column main effects against their respective stratum errors and for the interaction against the cell-level error. Reporting all three F-tests with their stratum-specific denominator df is the standard summary.

Interpretation

A reporting sentence: "The strip-plot ANOVA showed the row factor was marginally significant ($F_{2,4} = 3.8$, $p = 0.05$ against row-stratum error), the column factor was significant ($F_{3,6} = 6.4$, $p = 0.01$ against column-stratum error), and the row × column interaction was non-significant when tested against the cell-level error ($F_{6,24} = 1.2$, $p = 0.34$). Random-effect variance components for row-within-replicate and column-within-replicate were modest, indicating the strip structure absorbed limited but non-trivial nuisance variation." Always report each effect against its appropriate stratum.

Practical Tips

- Strip-plot designs require replication across blocks (multiple replicates of the cross structure) for valid inference; a single block leaves no error degrees of freedom for the row or column main effects.
- Strip-plot structure is most natural in agricultural field experiments where two perpendicular operations (e.g., ploughing direction and spraying direction) are applied independently, and in industrial manufacturing where two upstream processes cover whole batches in different ways.
- Correct random-effects specification is critical; an incorrect model treats restricted randomisation as if it were complete, yielding F-tests with the wrong denominator and inflated false-positive rates.
- Use `lmerTest::ranova()` to test the significance of the random-effect variance components themselves; this also helps diagnose whether the strip structure was substantively different from a CRD analysis would have implied.
- In industrial settings, strip-plot structure arises whenever two manufacturing stages each apply to a whole pallet, batch, or run before disaggregating; recognising the structure rather than ignoring it is essential to valid inference.
- For analysis software-checks, fit the same dataset with the wrong (CRD-like) model and the right (strip-plot) model; comparing the two highlights the inferential cost of ignoring the design and is a useful teaching diagnostic.

R Packages Used

`lme4::lmer()` and `lmerTest` for mixed-effects analysis with denominator-df adjustment; `agricolae::design.strip()` for the canonical strip-plot layout construction; `daewr` for textbook-companion strip-plot datasets and analysis helpers; `emmeans` for marginal means and contrasts at each stratum.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Observational designs and STROBE
- Randomised controlled trials
- Adaptive, non-inferiority, equivalence trials
- Bench and translational design
- MCAR, MAR, MNAR
- Multiple imputation with mice
- Linear mixed models with lme4
- GLMMs and GEE
- DAGs with dagitty and ggdag
- Propensity scores and IPTW
- G-methods, IV, DiD, RDD; heterogeneity of treatment effect
- SIR and SEIR models with deSolve
- Pre-registration and statistical analysis plans